NLP Assignment Collection

Two introductory NLP projects for data cleaning, text analysis, similarity search, and evidence-based reasoning

Overview for Students

These assignments ask you to use NLP as an investigation tool. You will work with messy real-world text, build structured datasets, extract useful signals, and present evidence-based conclusions. The goal is not only to run algorithms, but also to explain what the algorithms found and what they cannot prove.

General submission format

- A Colab with clean, readable code.
- A short report with figures and explanations.
- At least three visualizations for each assignment.
- A short discussion of limitations and possible errors.

Assignment 1: UFO/UAP Sightings Across Two Data Sources

Main question: Can we discover interesting UFO/UAP sightings and identify cases that appear in both a public civilian dataset and an official government dataset?

Background

In this assignment, you will analyze UFO/UAP sighting reports from two sources. The first source is a public Kaggle dataset of UFO sightings around the world. The second source is the official U.S. government PURSUE website, which contains UAP/UFO records released by war.gov. Your main task is to use NLP and similarity methods to find interesting patterns and, most importantly, to check whether some events were reported in both sources.

Data Sources

- Kaggle UFO Sightings Around the World: https://www.kaggle.com/datasets/camnugent/ufo-sightings-around-the-world
- Official PURSUE UAP/UFO records: https://www.war.gov/ufo/

Task	Points
Task 1: Data collection and unified schema	20
Task 2: NLP exploration of the reports	20
Task 3: Matching reports across the two sources	30
Task 4: Manual validation of candidate matches	15
Task 5: Summary, conclusions, and visualizations	15

Task 1: Data Collection and Unified Schema - 20 points

- Download and load the Kaggle UFO dataset.
- Collect metadata and documents from the war.gov PURSUE website. You may use the website table, downloadable files, or any metadata file provided by the site.
- Extract text from documents when possible.
- Create one unified table that can represent both sources.

Suggested unified fields:

```
source
record_id
date
city
state
country
latitude
longitude
location_text
description_text
object_shape
duration
source_file_or_link
```

Task 2: NLP Exploration - 20 points

- Find the most common words and phrases in the descriptions.
- Analyze common shapes and objects, such as light, sphere, triangle, disk, fireball, or formation.

- Use Named Entity Recognition to identify places, dates, organizations, military terms, and other important entities.
- Compare the language used in civilian reports with the language used in official records.
- Look for temporal trends, geographic trends, and unusual or rare sightings.
- Create visualizations that help explain the findings.

Task 3: Matching Reports Across the Two Sources - 30 points

This is the central task. Your goal is to find candidate pairs of reports that may describe the same event. Exact matching is not enough. Reports may use different wording, partial dates, or different names for the same location.

- Use several similarity signals:
  - Date similarity: same day, ±1 day, ±7 days, same month, or same year.
  - Location similarity: exact city/state, geographic distance, geocoding, or fuzzy matching of location names.
  - Text similarity: TF-IDF, cosine similarity, sentence embeddings, SBERT, spaCy vectors, or keyword overlap.
  - Entity similarity: similar places, dates, object shapes, colors, motion descriptions, aircraft, military bases, or weather terms.
- Use blocking to avoid comparing every report with every other report. For example, compare only reports that are close in date or location.
- Create a final matching score and explain the weights you chose.

Example scoring idea:

```
final_score =
    0.35 * text_similarity +
    0.25 * location_similarity +
    0.25 * date_similarity +
    0.15 * entity_similarity
```

For each top candidate match, report:

- Kaggle report ID and war.gov record ID or file name.
- Date and location in both sources.
- Short text from both reports.
- Similarity score.
- A short explanation of why the pair may describe the same event.

Task 4: Manual Validation - 15 points

- Manually review the top 20 candidate matches.
- Classify each pair as: likely same event, possibly same event, or probably not the same event.
- Discuss at least five examples in detail.
- Explain what made the matching difficult, such as missing dates, vague locations, generic descriptions, redacted records, or different writing styles.

Task 5: Summary and Insights - 15 points

- What are the most interesting UFO/UAP patterns you found?
- Did you find cases that likely appear in both datasets?
- Which NLP methods worked best?
- Which methods failed or created false matches?
- What would you improve with more time?

Suggested Tools

- pandas, numpy, scikit-learn
- spaCy or NLTK
- sentence-transformers
- rapidfuzz
- geopy
- Plotly Express or matplotlib
- PyMuPDF or pdfplumber

Assignment 2: Who Killed Edwin Drood?

Main question: Can NLP help us build a data-based theory about the unfinished mystery in Charles Dickens' final novel?

Background

The Mystery of Edwin Drood is the last novel written by Charles Dickens. Dickens died in 1870 before he finished the book, so the main mystery was never officially solved. The story is about Edwin Drood, a young man who suddenly disappears. Many readers believe he was murdered, but the book does not give a final answer.

The main questions are:

- Who most likely killed Edwin Drood?
- Was Edwin Drood really murdered?
- Which textual clues support each theory?

Before starting, it is recommended to listen to Doron Fishler's podcast episode: מי רצח את אדווין דרוד? חלק א׳ or read about the book and the back story.

The podcast gives useful background, but your answer must be based on your own NLP analysis of the text.

Data Source

Use the full text of *The Mystery of Edwin Drood*. Recommended source: Project Gutenberg eBook 564

Task	Points
Task 1: Prepare the text	15
Task 2: Character analysis	20
Task 3: Finding suspects	25
Task 4: Extract important clues	20
Task 5: Compare to previous works by Dickens	10
Task 6: Final conclusion	10

Task 1: Prepare the Text - 15 points

- Download the novel text and remove Project Gutenberg headers and footers.
- Split the novel into chapters.
- Split the chapters into paragraphs and/or sentences.
- Clean the text and create a structured table.

Suggested table fields:

```
chapter
paragraph_id
sentence_id
text
characters_mentioned
```

Track the main characters, including:

```
Edwin Drood
John Jasper
Rosa Bud
Neville Landless
```

```
Helena Landless
Mr. Crisparkle
Durdles
Princess Puffer
Dick Datchery
```

Task 2: Character Analysis - 20 points

- Count how many times each character is mentioned.
- Show in which chapters each character appears.
- Find which characters appear together in the same paragraph, scene, or chapter.
- Find words that often appear near each character.
- Analyze whether the language around each character is positive, negative, dark, suspicious, emotional, or violent.

Suggested visualizations:

- Character mention frequency.
- Character mentions by chapter.
- Character co-occurrence network.
- Sentiment around each main character.

Task 3: Finding Suspects - 25 points

Choose several possible suspects and analyze them using NLP. For each suspect, look for evidence related to motive, opportunity, and suspicious language.

- Motive: Does the character have a reason to hurt Edwin?
    - Examples: jealousy, love, anger, rivalry, revenge, fear, obsession.
- Opportunity: Was the character close to Edwin before he disappeared?
    - Examples: same chapter, same scene, mentioned near Edwin, connected to the location of the disappearance.
- Suspicious language: Does the text around the character include suspicious themes?
    - Examples: death, murder, fear, darkness, secret, opium, jealousy, graveyard, disappearance, violence.

Example scoring idea:

```
suspicion_score =
    motive_score +
    opportunity_score +
    suspicious_language_score
```

Task 4: Extract Important Clues - 20 points

- Use TF-IDF, keyword extraction, Named Entity Recognition, topic modeling, sentence embeddings, or sentiment analysis.
- Search for suspicious words and repeated motifs.
- Cluster similar paragraphs or scenes.
- Find at least 8-10 important clues.

For each clue, show:

```
chapter
short quote or sentence
characters involved
why this clue is important
which suspect it supports
```

Task 5: Compare to Previous Works by Dickens - 10 points

Instead of comparing only to detective novels by other authors, compare the plot line and themes of *The Mystery of Edwin Drood* to earlier works by Dickens. The goal is to ask whether Dickens reused patterns from his previous novels, and whether those patterns help us guess the likely direction of the unfinished story.

Possible Dickens works to compare:

- Oliver Twist
- Bleak House
- David Copperfield
- Great Expectations
- Our Mutual Friend
- A Tale of Two Cities

Things to compare:

- Plot arc: how the story opens, when secrets appear, and when revelations usually happen.
- Character roles: villain, victim, false suspect, hidden helper, orphan or young hero, mysterious stranger.
- Common Dickens themes: secret identities, guilt, obsession, social pressure, hidden pasts, doubles, disguise, confession, and delayed justice.
- Language patterns: dark or gothic words, emotional intensity, moral judgment, fear, crime, and secrecy.
- Character networks: whether the villain is socially close to the victim or hidden at the edge of the story.

Possible NLP approaches:

- Compare chapter-level sentiment arcs across novels.
- Use topic modeling to compare themes across books.
- Create embeddings for chapter summaries and compare plot progression.
- Build character co-occurrence networks for each novel and compare their structures.
- Search for repeated motifs such as secrecy, guilt, disappearance, inheritance, obsession, and confession.

Final question for this task: **Does Edwin Drood look like a continuation of patterns Dickens used before, and do those patterns support one suspect or one theory more strongly?**

Task 6: Final Conclusion - 10 points

- State your main suspect or theory.
- Give a confidence level.
- Present the strongest clues supporting your theory.
- Discuss other possible explanations.
- Explain what your NLP analysis found and what it cannot prove.

Suggested Tools

- pandas, spaCy, NLTK
- scikit-learn
- sentence-transformers
- NetworkX
- Plotly Express or matplotlib
- BERTopic or another topic modeling method

Final Submission

- A colab Notebook.
- A short report, 2-4 pages.
- At least three visualizations.
- A ranked list of suspects.
- A table of important clues.
- A short comparison to at least one previous work by Dickens.

Important Note

This exercise is not about finding the certain truth. Dickens never completed the story. The goal is to use NLP to support a literary investigation and show how computational tools can help us analyze characters, clues, and plot structure.

Source Links

- Kaggle UFO Sightings Around the World: https://www.kaggle.com/datasets/camnugent/ufo-sightings-around-the-world
- war.gov PURSUE UAP/UFO records: https://www.war.gov/ufo/
- Project Gutenberg: The Mystery of Edwin Drood: https://www.gutenberg.org/ebooks/564
- Doron Fishler podcast: מי רצח את אדווין דרוד? חלק א׳: https://doronfishler.com/episodes/מי-רצח-את-אדווין-דרוד-חלק-א/